

WIPRO CAPSTONE PROJECT REPORT

Name	Kiran Sakthivel A
Superset ID	5304127
Organization	Wipro
Project Type	Capstone Project
Project Name	Automation Testing using Selenium PyTest Framework Automation Testing using Behave BDD Framework
Domain	Healthcare
Module Title	Apollo 24/7 – Lab Test Module Automation

SYNOPSIS:

The Wipro Capstone Project focuses on automating the Apollo 24/7 Lab Tests module using two separate automation frameworks: Selenium WebDriver with Python PyTest framework and Behave BDD framework. The automation validates lab test search, health package selection, cart operations, patient details entry, and the checkout workflow.

The framework was designed using the Page Object Model (POM) architecture to improve maintainability and reusability. Screenshots, logs, reports, and reusable utilities were integrated to support execution tracking and debugging.

The project includes both positive and negative testing scenarios using industry-standard automation testing practices.

PROJECT DESCRIPTION:

The automation framework automates the end-to-end workflow of the Apollo 24/7 Lab Tests module. The framework performs popular health checkup selection, dynamically verifying critical application features such as package card visibility, dynamic pricing display, and the complete checkout flow.

Separate page classes were implemented for Home, Lab Tests, Package Details, Cart, and Checkout modules using Selenium Python Page Object Model framework.

The project also includes Behave BDD implementation where business scenarios are written in Gherkin syntax and integrated with Selenium automation scripts.

SELENIUM PYTEST FRAMEWORK:

TEST STRATEGY:

The project uses Selenium WebDriver with Python PyTest framework for automation testing. The framework supports:

- End-to-End Testing
- Positive Testing
- Negative Testing
- Data-Driven Testing

Tools Used:

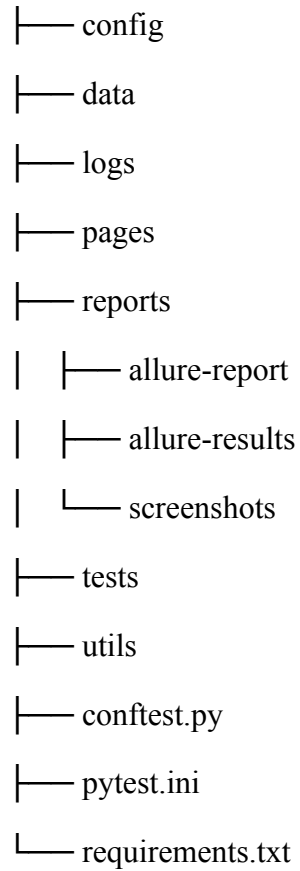
- Selenium WebDriver
- PyTest
- Allure Reports
- Logging Module

TEST ENVIRONMENT:

Operating System	Windows 11
IDE	PyCharm
Browser	Edge
Framework	Selenium WebDriver + PyTest
Design Pattern	Page Object Model (POM)
Reporting Tool	Allure Reports

FRAMEWORK ARCHITECTURE:

Apollo247Automation



TEST EXECUTION:

The automation suite includes:

- 1 End-to-End Test Case
- 4 Positive Test Cases
- 4 Negative Test Cases

The End-to-End test validates the complete Apollo 24/7 checkout workflow from lab tests navigation until checkout verification.

SELENIUM TEST CASE SUMMARY:

S.No	Test Type	Test Scenario	Expected Result
1	End-to-End	Complete Checkout Workflow	User successfully reaches checkout
2	Positive	Popular Packages Page Load	Packages loaded successfully
3	Positive	Apply Category Filter	Category filter applied successfully
4	Positive	Sort High to Low	Packages sorted by price high to low
5	Positive	View Package Details	Product Description Page opens
6	Negative	Invalid Login	OTP screen does not appear for invalid number
7	Negative	Invalid URL Category	Page handles invalid category gracefully

8	Negative	Duplicate Add to Cart	Duplicate addition is handled gracefully
9	Negative	Proceed Without Patient	Error displayed if no patient selected

END-TO-END FLOW VALIDATION:

The End-to-End automation validates the complete business workflow including:

- User Login
- Lab Tests Navigation
- Package Selection
- Cart Verification
- Patient Details Entry
- Checkout Verification

POSITIVE TEST CASES

Positive Test Case 1 – Popular Packages Page Load

- Validates successful navigation to Popular Health Checkups
- Verifies page loaded successfully with packages

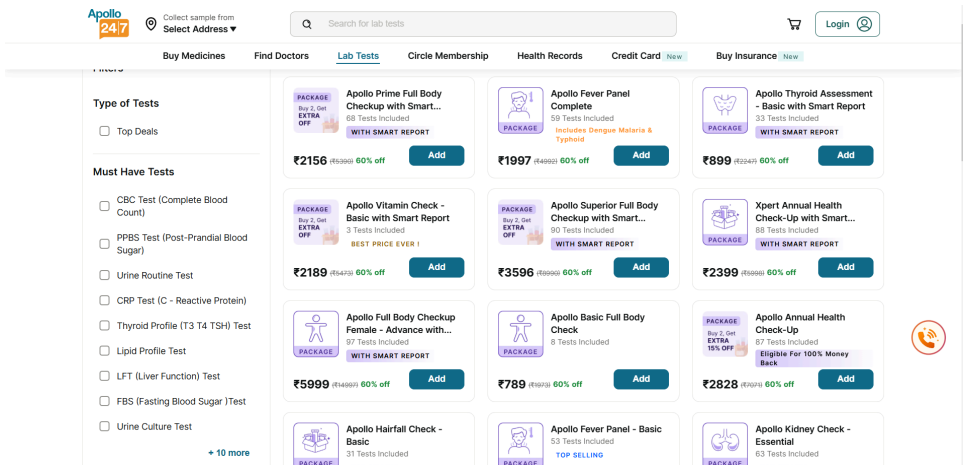


Fig: Popular Packages Page Load

Positive Test Case 2 – Apply Category Filter

- Selects specific health filters (e.g. Women's Health)
- Verifies filter application successfully

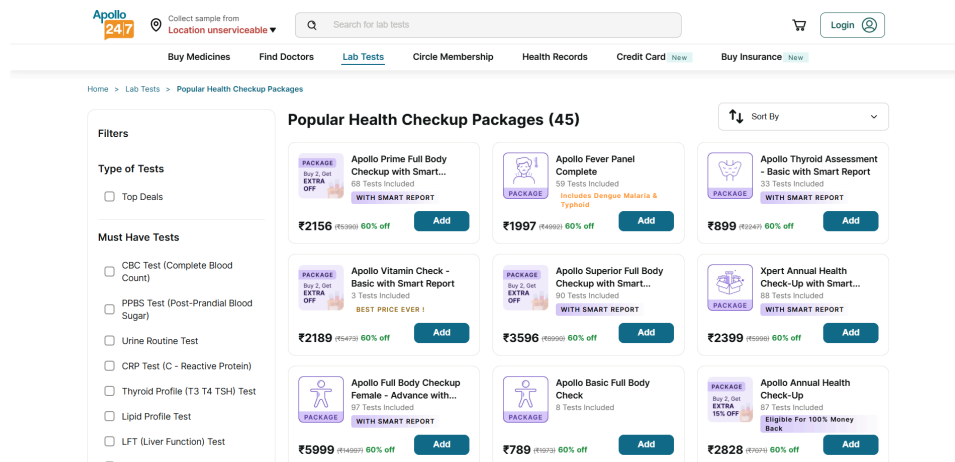


Fig: Category Filter Applied

Positive Test Case 3 – Sort High to Low

- Selects 'Price: High to Low' from Sort dropdown
- Verifies sorting interaction

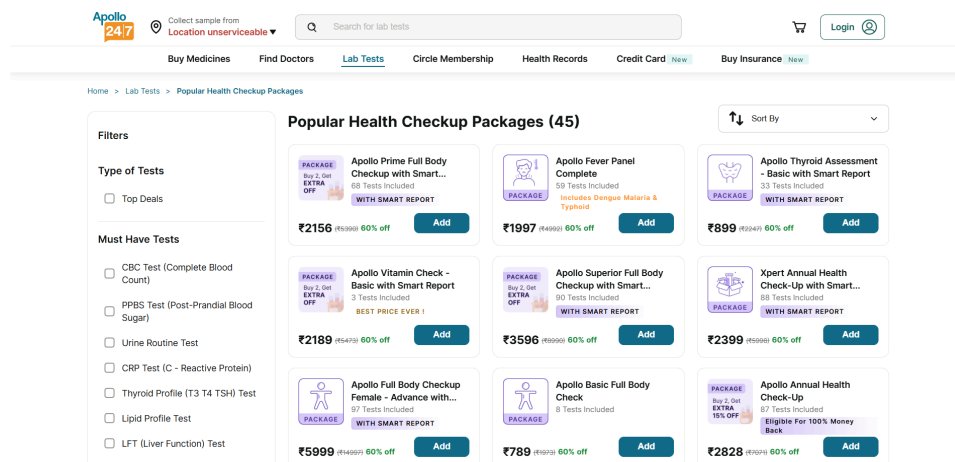


Fig: Packages Sorted High to Low

Positive Test Case 4 – View Package Details

- Clicks on a package title
- Verifies Product Description Page loaded

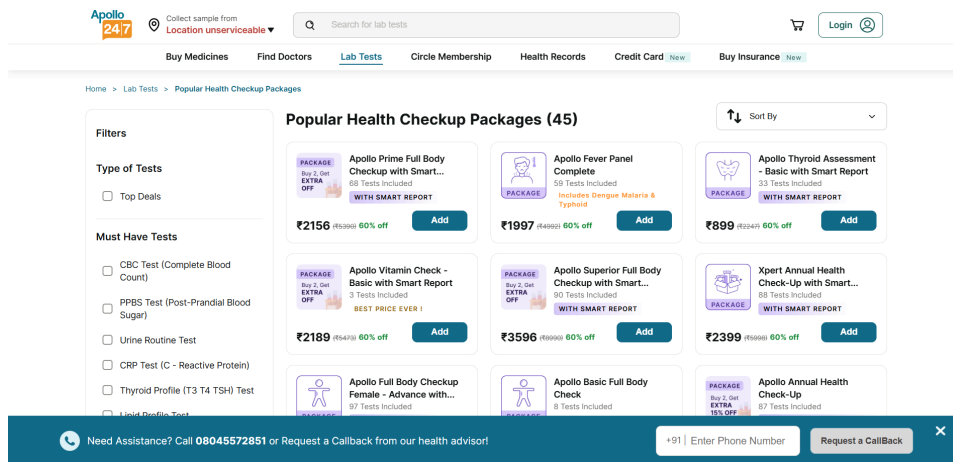


Fig: Product Description Page

NEGATIVE TEST CASES:

Negative Test Case 1 – Invalid Login

- Enters invalid mobile number
- Validates OTP screen does not appear

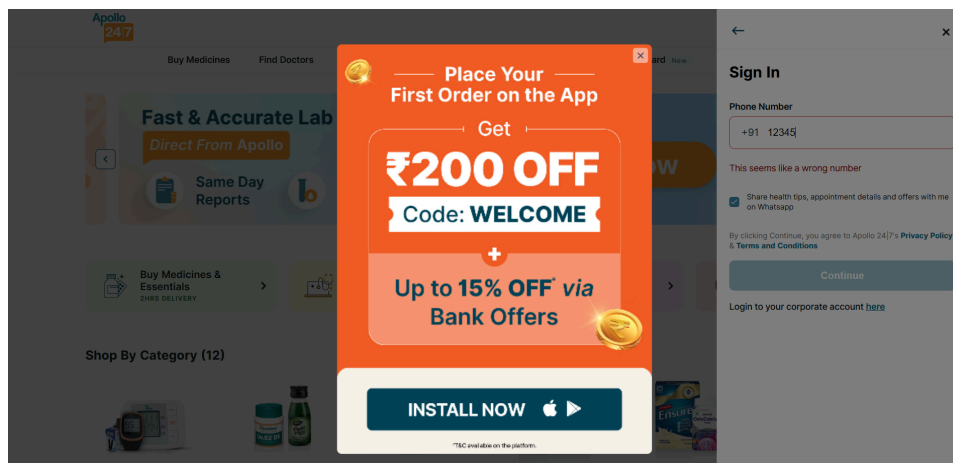


Fig: Invalid Login Handling

Negative Test Case 2 – Invalid URL Category

- Navigates to Popular Health Checkups with bogus category
- Verifies page handles invalid category gracefully

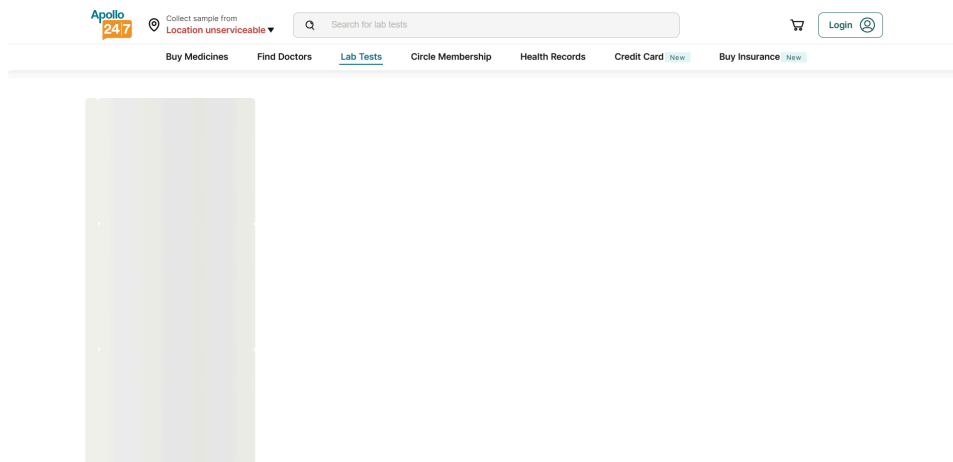


Fig: Invalid URL Category Handled

Negative Test Case 3 – Duplicate Add to Cart

- Attempts to add same package again from listing
- Verifies duplicate addition is handled gracefully

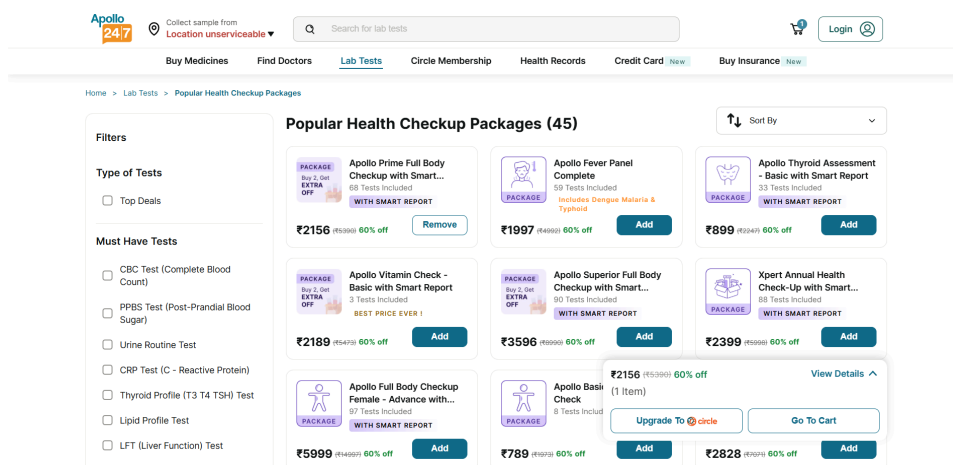


Fig: Duplicate Add to Cart Graceful Handling

Negative Test Case 4 – Proceed Without Patient

- Attempts to proceed from cart without selecting patient
- Verifies error or remains on patient screen

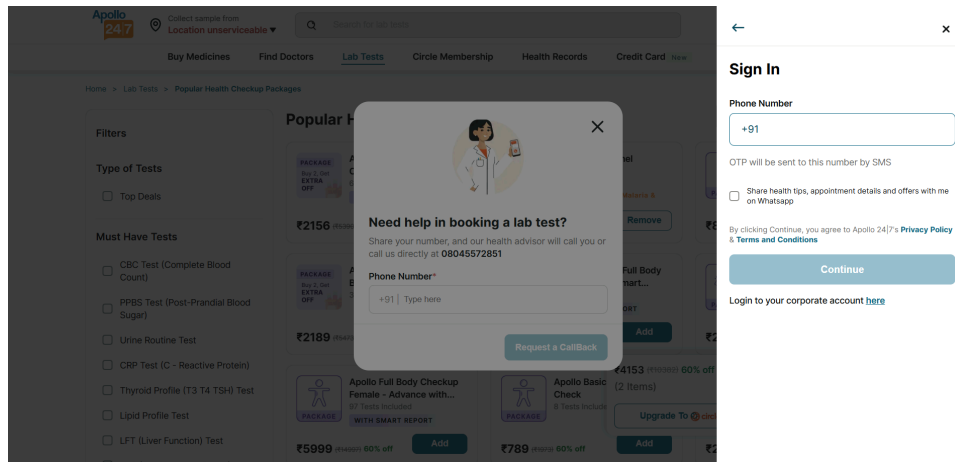


Fig: Proceed Without Patient Blocked

ALLURE REPORTS:

- Integrated Allure Reports with PyTest
- Generates professional execution reports
- Displays passed and failed test status
- Captures screenshots during execution
- Maintains execution history
- Displays step-by-step execution flow
- Helps debugging and tracking failures

TEST OUTPUTS:

The framework automatically captures screenshots during important execution stages including login page, lab tests page, package selection, and cart verification.

Detailed execution logs are generated using Python logging framework for debugging and execution tracking.

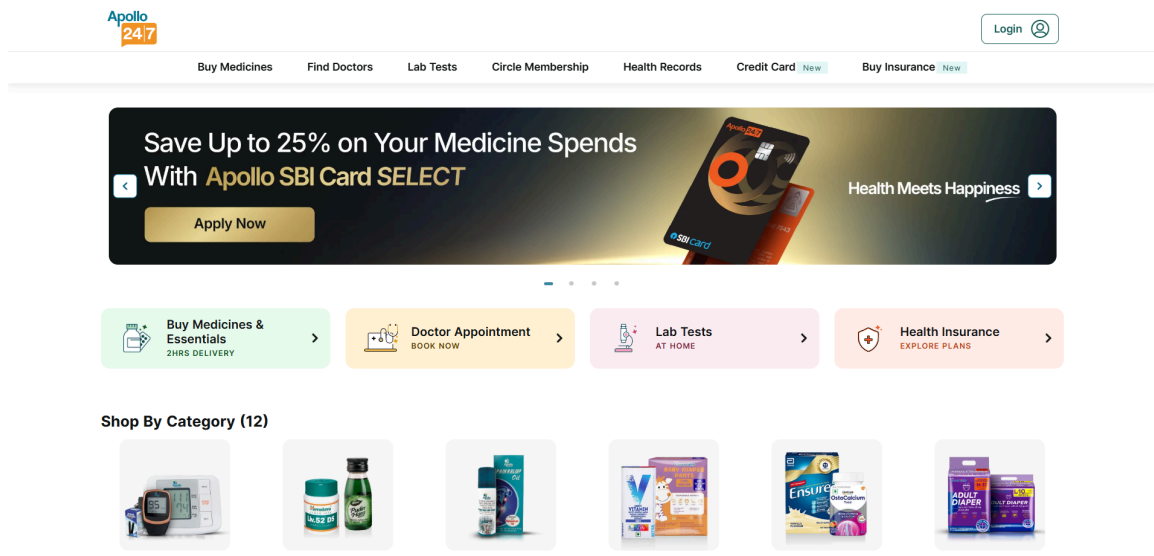


Fig 1: 01_homepage

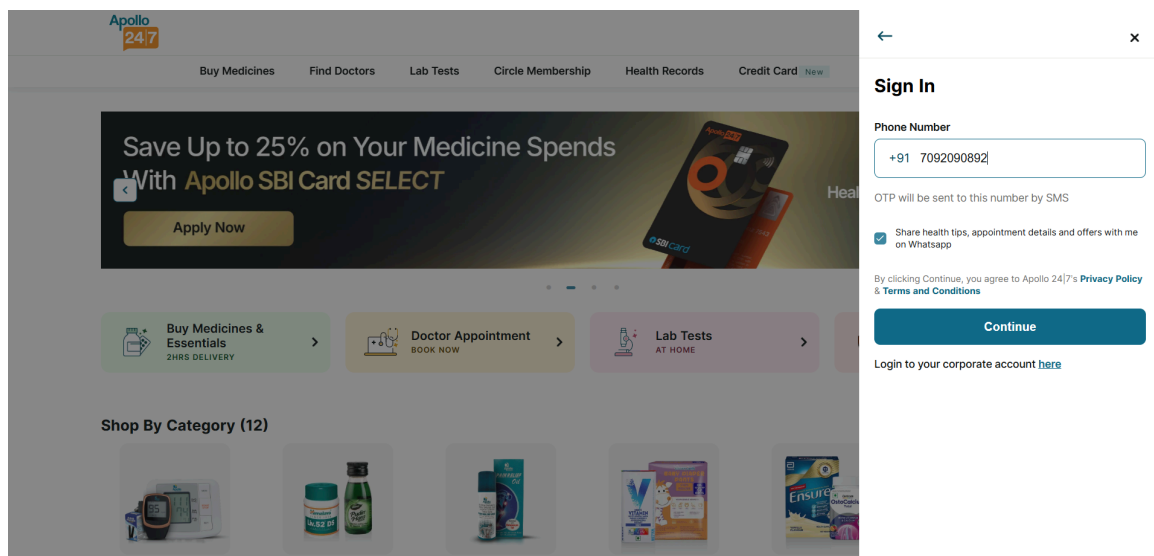


Fig 2: 02_mobile_entered

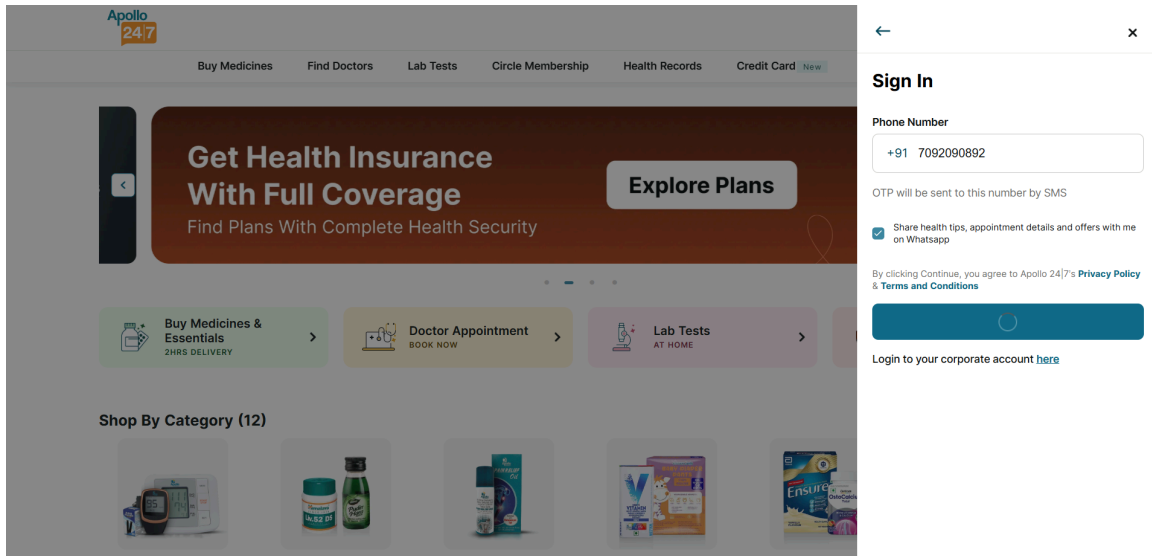


Fig 3: 03_otp_screen

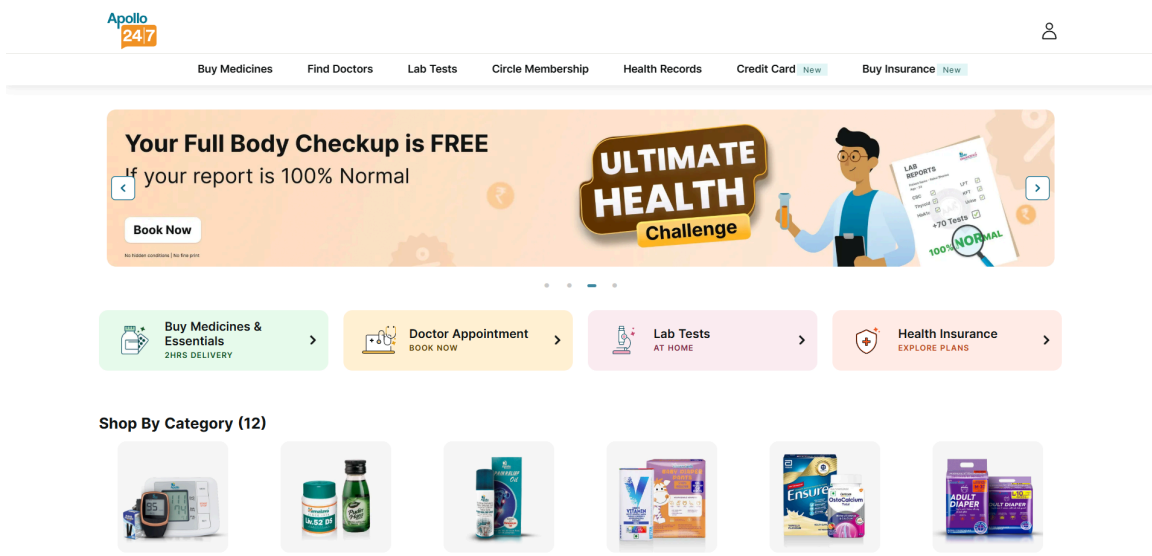


Fig 4: 04_after_login

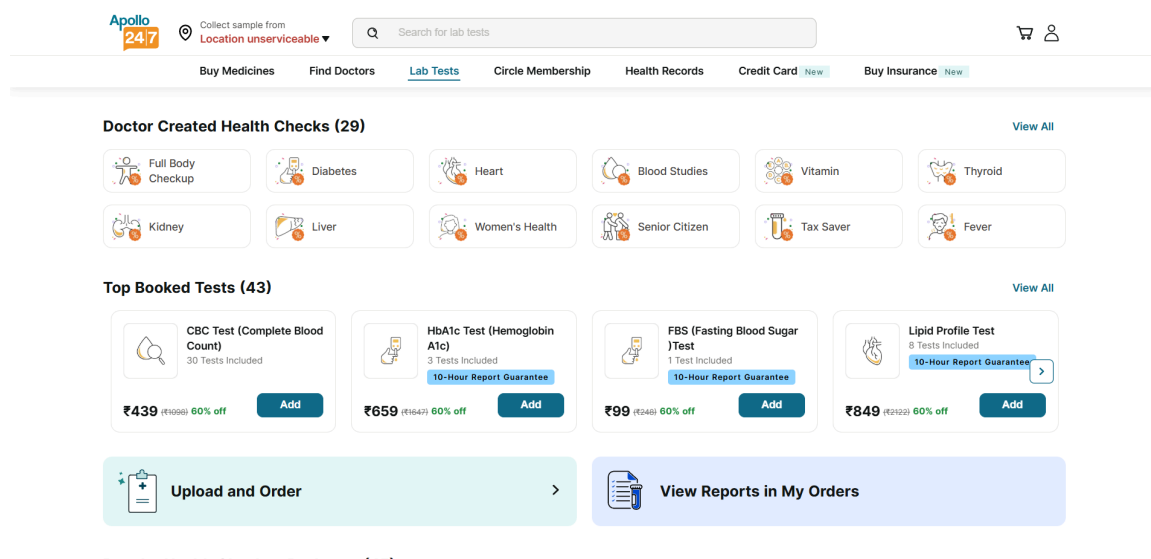


Fig 5: 05_lab_tests_page

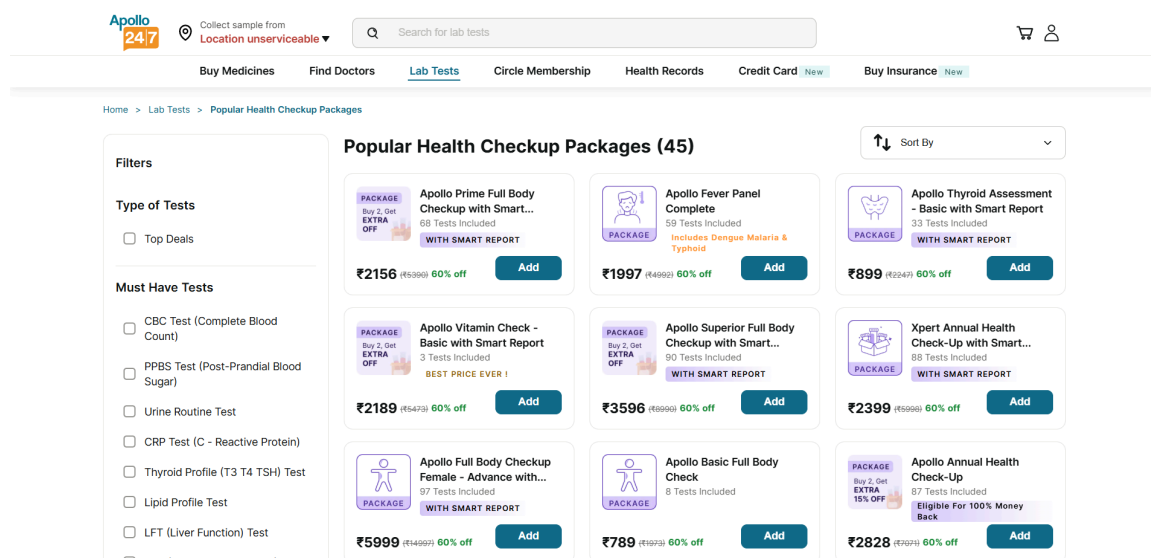


Fig 6: 06_popular_packages_listing

Collect sample from
Location unserviceable

Search for lab tests

Buy Medicines
Find Doctors
Lab Tests
Circle Membership
Health Records
Credit Card
Buy Insurance

Home > Lab Tests > Popular Health Checkup Packages

Filters

Clear All

Type of Tests

☒ Top Deals

Must Have Tests

☐ CBC Test (Complete Blood Count)
☐ PPBS Test (Post-Prandial Blood Sugar)
☐ Urine Routine Test
☐ CRP Test (C - Reactive Protein)
☐ Thyroid Profile (T3 T4 TSH) Test
☐ Lipid Profile Test
☐ LFT (Liver Function) Test
☐ EBC (Eating Blood Count) Test

Popular Health Checkup Packages (45)

Sort By

Apollo Hairfall Check Advance Female

60 Tests Included

VALUE FOR MONEY

₹6249

(₹15622) 60% off

Add

Apollo Liver Check - Advance

75 Tests Included

VALUE FOR MONEY

₹6009

(₹15023) 60% off

Add

Apollo Full Body Checkup Female - Advance with...

97 Tests Included

WITH SMART REPORT

₹5999

(₹14997) 60% off

Add

Apollo Fever Panel - Advance

67 Tests Included

APOLLO'S CHOICE

₹4949

(₹12372) 60% off

Add

Apollo Full Body Checkup

90 Tests Included

₹3455

(₹8629) 65% off

Add

Apollo Hairfall Check Advance Male

54 Tests Included

VALUE FOR MONEY

₹4029

(₹10072) 60% off

Add

Quadruple Marker Test

31 Tests Included

₹3975

(₹9938) 60% off

Add

Apollo Thyroid Assessment - Advance

37 Tests Included

APOLLO'S CHOICE

₹3881

(₹9702) 60% off

Add

Apollo Vitamin Check - Advance with Smart Report

41 Tests Included

WITH SMART REPORT

₹3819

(₹9548) 60% off

Add

Fig 7: 07_filter_top_deals

Collect sample from
Location unserviceable

Search for lab tests

Buy Medicines
Find Doctors
Lab Tests
Circle Membership
Health Records
Credit Card
Buy Insurance

Home > Lab Tests > Popular Health Checkup Packages

Filters

Clear All

Type of Tests

☒ Top Deals

Must Have Tests

☐ CBC Test (Complete Blood Count)
☐ PPBS Test (Post-Prandial Blood Sugar)
☐ Urine Routine Test
☐ CRP Test (C - Reactive Protein)
☐ Thyroid Profile (T3 T4 TSH) Test
☐ Lipid Profile Test
☐ LFT (Liver Function) Test
☐ EBC (Eating Blood Count) Test

Popular Health Checkup Packages (9)

Sort By

Apollo Full Body Checkup

90 Tests Included

₹3455

(₹8629) 65% off

Add

Apollo Superior Full Body Checkup with Smart...

90 Tests Included

WITH SMART REPORT

₹3596

(₹8999) 60% off

Add

Apollo Annual Health Check-Up

87 Tests Included

Eligible For 100% Money Back

₹2828

(₹7079) 60% off

Add

Apollo Full Body Check - Advance I

79 Tests Included

TAX SAVER

₹2719

(₹6791) 60% off

Add

Xpert Annual Health Check-Up with Smart...

88 Tests Included

WITH SMART REPORT

₹2399

(₹5998) 60% off

Add

Apollo Prime Full Body Checkup with Smart...

68 Tests Included

WITH SMART REPORT

₹2156

(₹5390) 60% off

Add

Apollo Vitamin Check (Vit D + Vit B12) with Smart...

2 Tests Included

WITH SMART REPORT

₹1823

(₹4452) 60% off

Add

Apollo Full Body Check - Basic I

59 Tests Included

₹1419

(₹3540) 60% off

Add

Apollo Full Body Checkup - Basic

52 Tests Included

₹879

(₹2198) 60% off

Add

Fig 8: 08_filter_mens_health

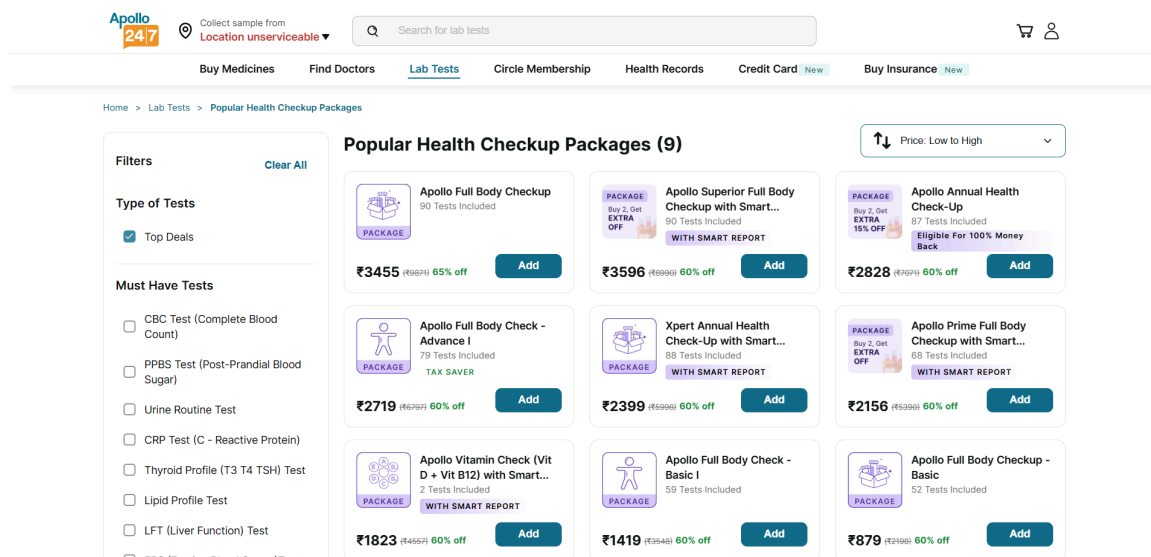


Fig 9: 08_sort_low_to_high

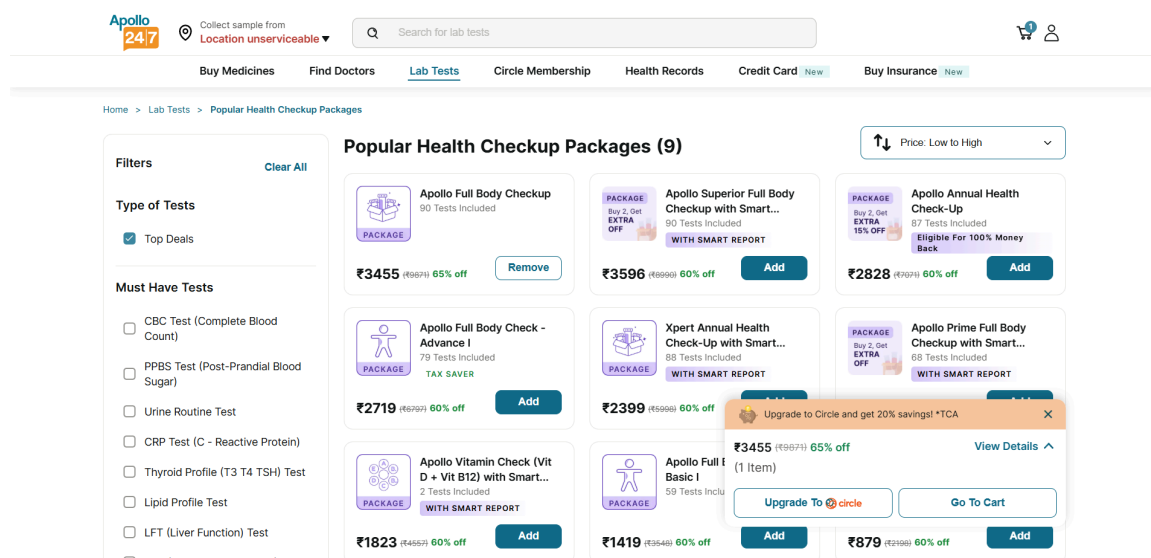


Fig 10: 09_package_added

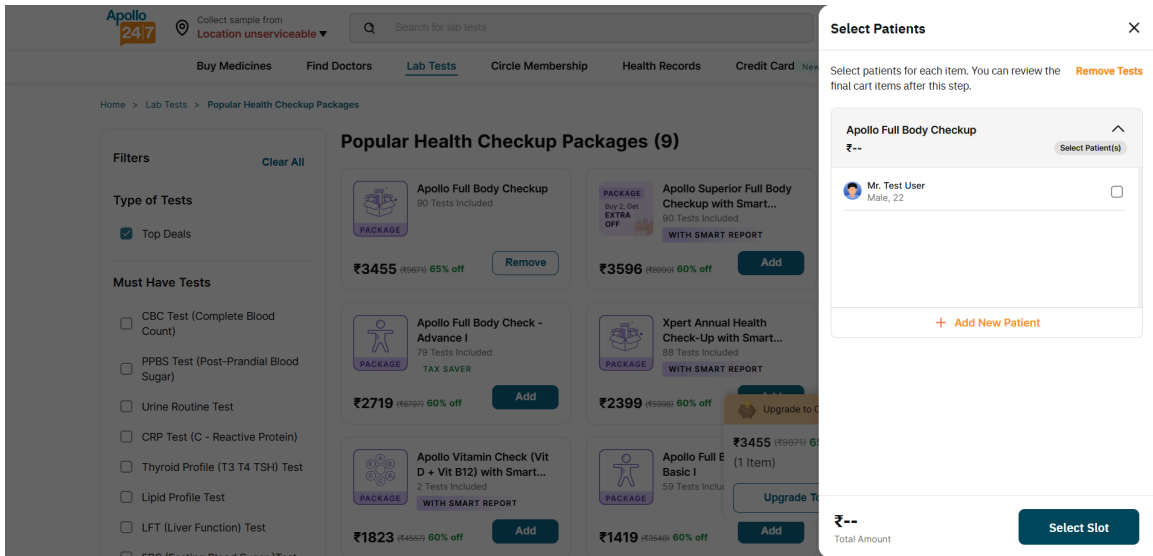


Fig 11: 10_cart_popup

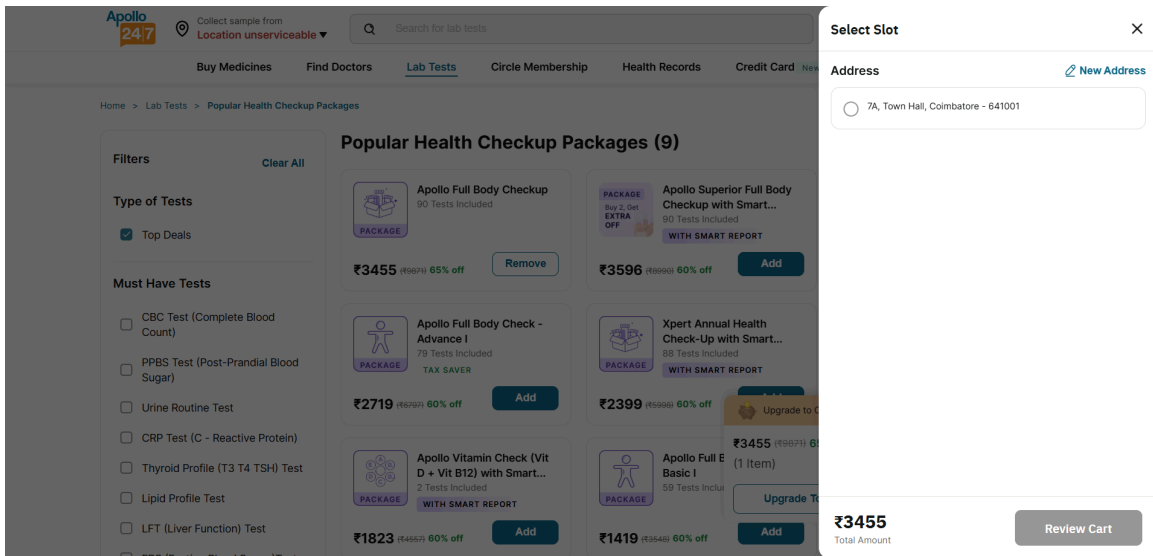


Fig 12: 11_patient_selected

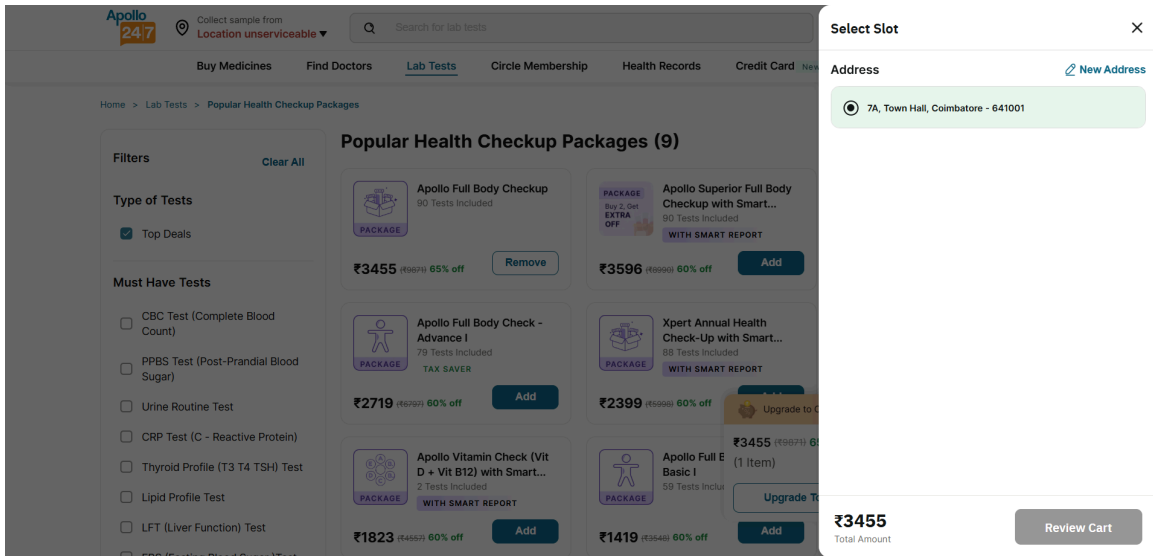


Fig 13: 12_address_selected

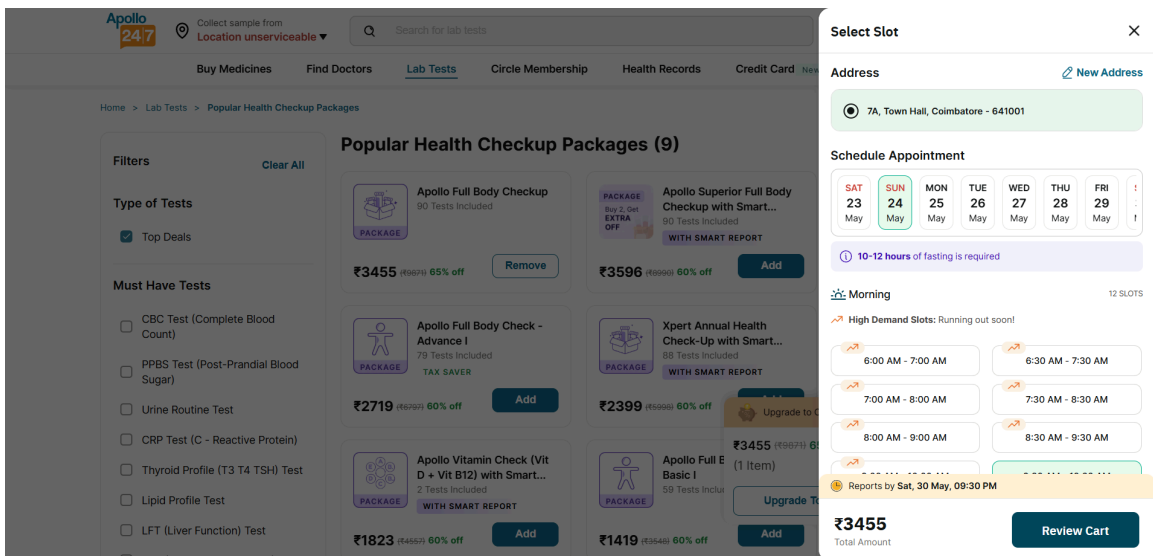


Fig 14: 13_schedule_selected

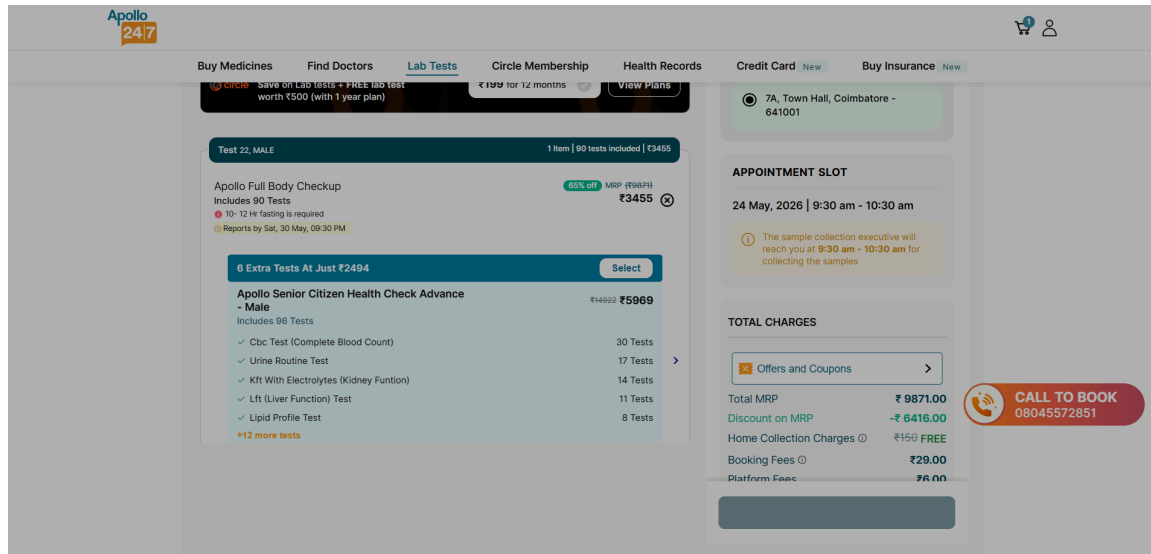


Fig 15: 14_proceed_to_pay

TEST REPORT:

- Total Tests Executed: 9
- Tests Passed: 9
- Tests Failed: 0
- Automation Accuracy: 100%

CONCLUSION:

The Selenium PyTest automation framework successfully validates the Apollo 24/7 Lab Tests application using End-to-End, Positive and Negative testing approaches.

The framework is designed using:

- Selenium WebDriver
- PyTest Framework
- Page Object Model (POM)
- Allure Reporting
- Logging Framework
- Data-Driven Testing

The automation framework improves:

- Test maintainability
- Reusability
- Scalability
- Reporting quality
- Automation reliability

The framework successfully validates critical business workflows including lab tests navigation, filter validation, package selection, cart operations and checkout verification.

All test cases executed successfully with professional reporting, screenshot capture and detailed execution logs.

Overall, the project demonstrates a practical and professional Selenium automation framework for real-world web application testing.

SELENIUM BDD FRAMEWORK:

TEST STRATEGY:

The project uses Selenium WebDriver with Python Behave BDD framework for automation testing.

- End-to-End Testing
- Positive Testing
- Negative Testing
- Data-Driven Testing
- Behavior-Driven Development (BDD)

Tools Used:

- Selenium WebDriver
- Behave
- Allure Reports
- Logging Module

TEST ENVIRONMENT:

Operating System	Windows 10 / Windows 11
IDE	VS Code / PyCharm
Browser	Google Chrome (or) Edge
Framework	Selenium WebDriver + Behave
Design Pattern	Page Object Model (POM)

Reporting Tool	Allure Reports
Test Approach	Behavior Driven Development

FRAMEWORK ARCHITECTURE:

Apollo247BDDAutomation

- |— config
- |— data
- |— features
 - | |— steps
 - | |— environment.py
 - | |— *.feature
- |— locators
- |— logs
- |— pages
- |— reports
 - | |— allure-report
 - | |— allure-results
 - | |— screenshots
- |— utils
- |— behave.ini
- |— pytest.ini
- |— runtest.py
- |— requirements.txt

BDD IMPLEMENTATION:

The framework follows Behavior Driven Development using:

- Feature Files
- Step Definition Files
- Gherkin Syntax
- Given-When-Then Structure

TEST EXECUTION:

The BDD automation suite includes:

- 1 End-to-End Test Case
- 4 Positive Test Cases
- 4 Negative Test Cases

The End-to-End scenario validates the complete Apollo 24/7 checkout workflow.

BDD TEST CASE SUMMARY:

S.No	Test Type	Test Scenario	Expected Result
1	End-to-End	Complete Checkout Workflow	User successfully reaches checkout
2	Positive	Popular Packages Page Load	Packages loaded successfully

3	Positive	Apply Category Filter	Category filter applied successfully
4	Positive	Sort High to Low	Packages sorted by price high to low
5	Positive	View Package Details	Product Description Page opens
6	Negative	Invalid Login	OTP screen does not appear for invalid number
7	Negative	Invalid URL Category	Page handles invalid category gracefully
8	Negative	Duplicate Add to Cart	Duplicate addition is handled gracefully
9	Negative	Proceed Without Patient	Error displayed if no patient selected

END-TO-END FLOW VALIDATION:

The End-to-End automation validates the complete business workflow including:

- User Login
- Lab Tests Navigation

- Package Selection
- Cart Verification
- Patient Details Entry
- Checkout Verification

POSITIVE TEST CASES:

Positive Test Case 1 – Popular Packages Page Load

- Validates successful navigation to Popular Health Checkups
- Verifies page loaded successfully with packages

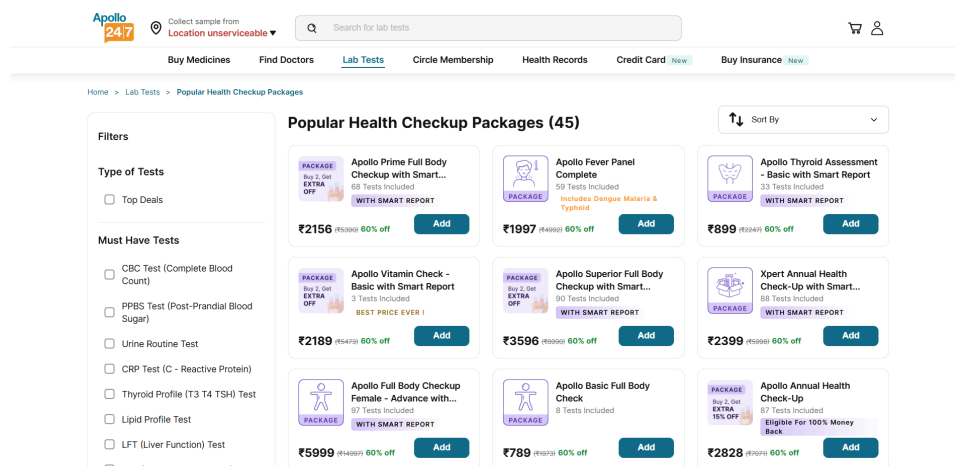


Fig: Popular Packages Page Load

Positive Test Case 2 – Apply Category Filter

- Selects specific health filters (e.g. Women's Health)
- Verifies filter application successfully

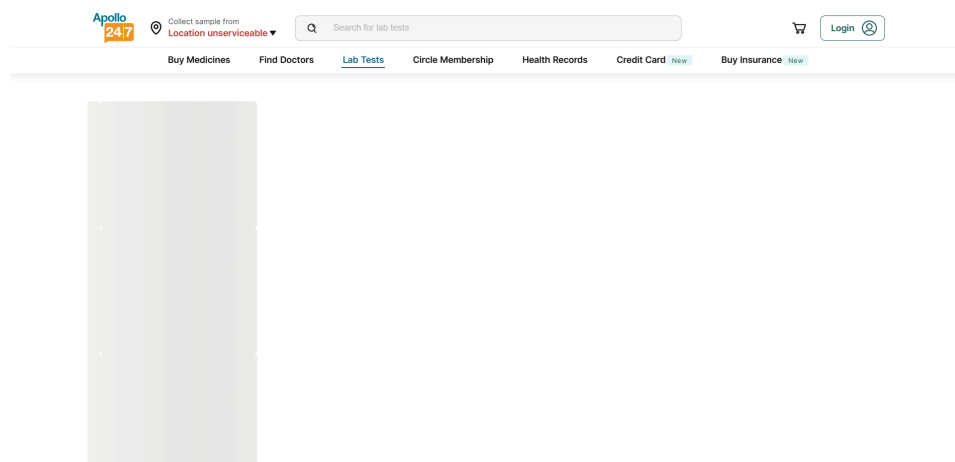


Fig: Category Filter Applied

Positive Test Case 3 – Sort High to Low

- Selects 'Price: High to Low' from Sort dropdown
- Verifies sorting interaction

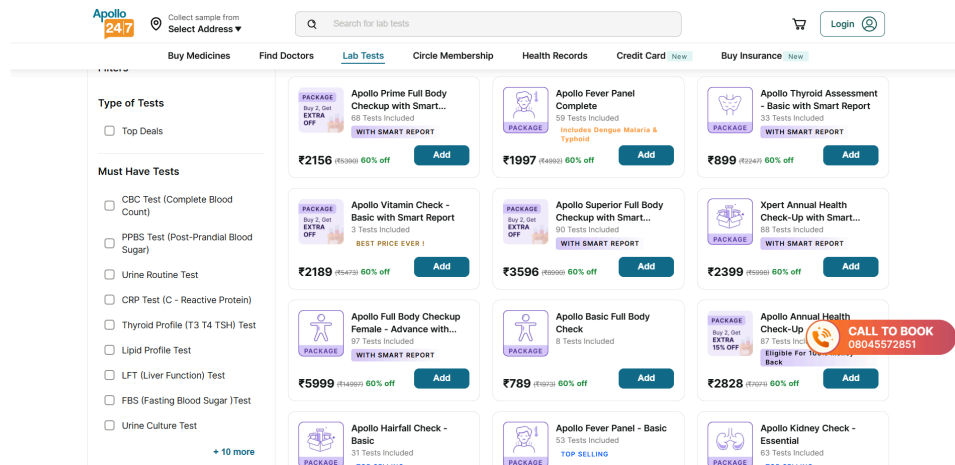


Fig: Packages Sorted High to Low

Positive Test Case 4 – View Package Details

- Clicks on a package title
- Verifies Product Description Page loaded

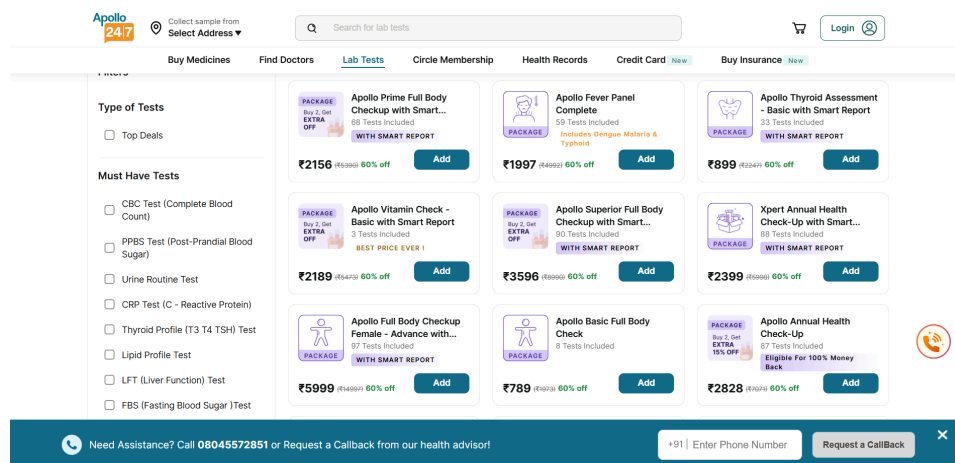


Fig: Product Description Page

NEGATIVE TEST CASES:

Negative Test Case 1 – Invalid Login

- Enters invalid mobile number
- Validates OTP screen does not appear

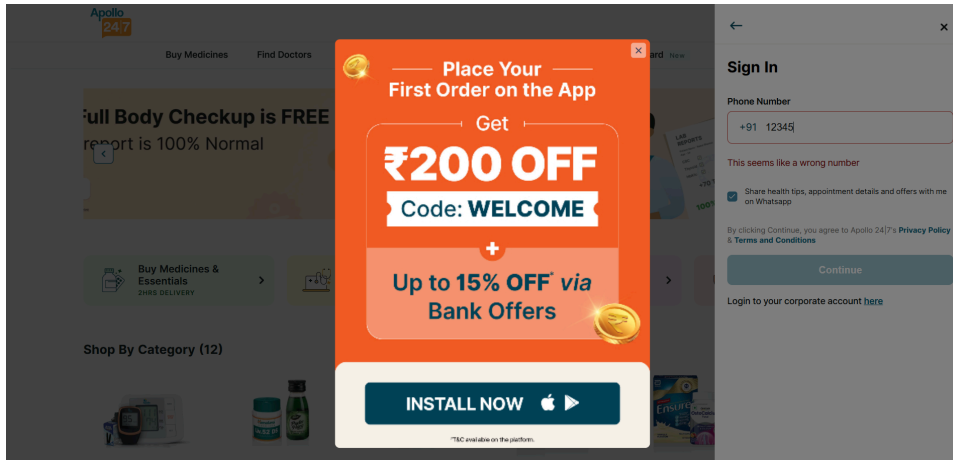


Fig: Invalid Login Handling

Negative Test Case 2 – Invalid URL Category

- Navigates to Popular Health Checkups with bogus category
- Verifies page handles invalid category gracefully

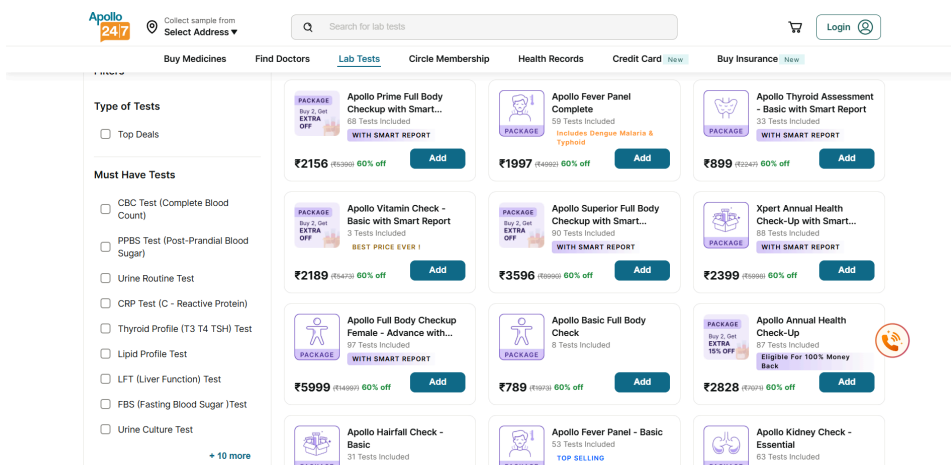


Fig: Invalid URL Category Handled

Negative Test Case 3 – Duplicate Add to Cart

- Attempts to add same package again from listing
- Verifies duplicate addition is handled gracefully

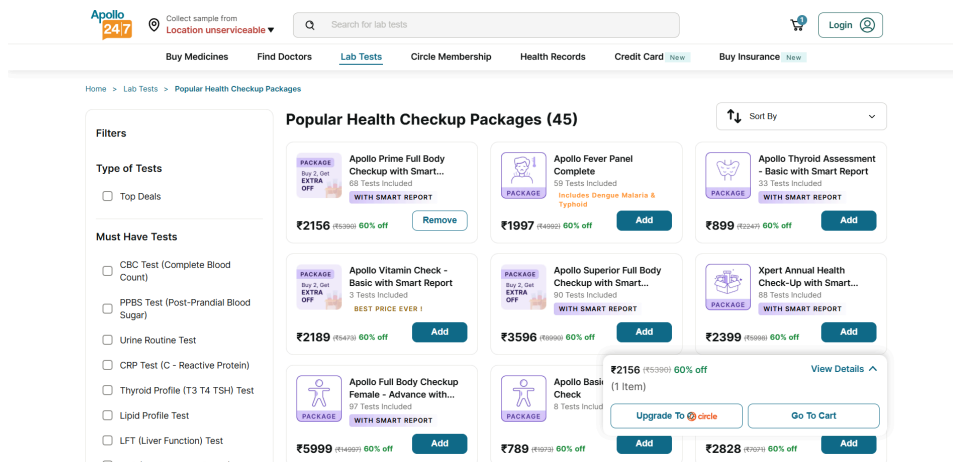


Fig: Duplicate Add to Cart Graceful Handling

Negative Test Case 4 – Proceed Without Patient

- Attempts to proceed from cart without selecting patient
- Verifies error or remains on patient screen

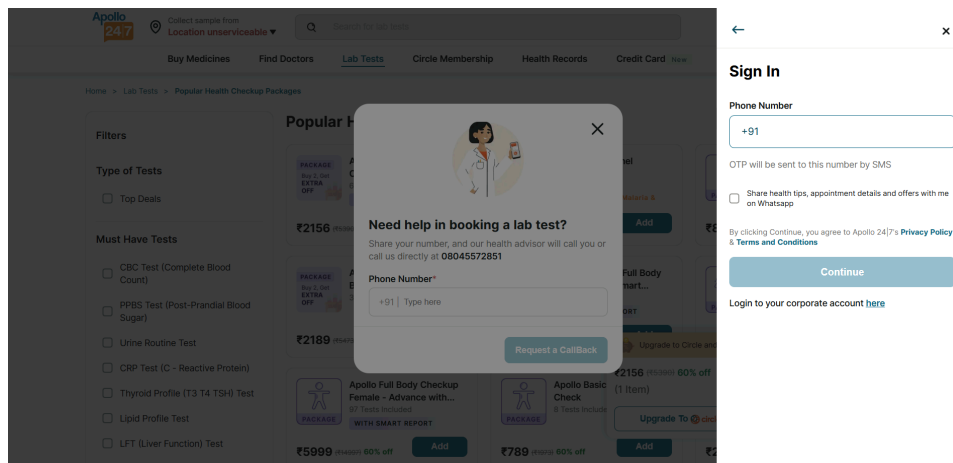


Fig: Proceed Without Patient Blocked

TEST OUTPUTS:

The BDD framework automatically captures screenshots during step executions.

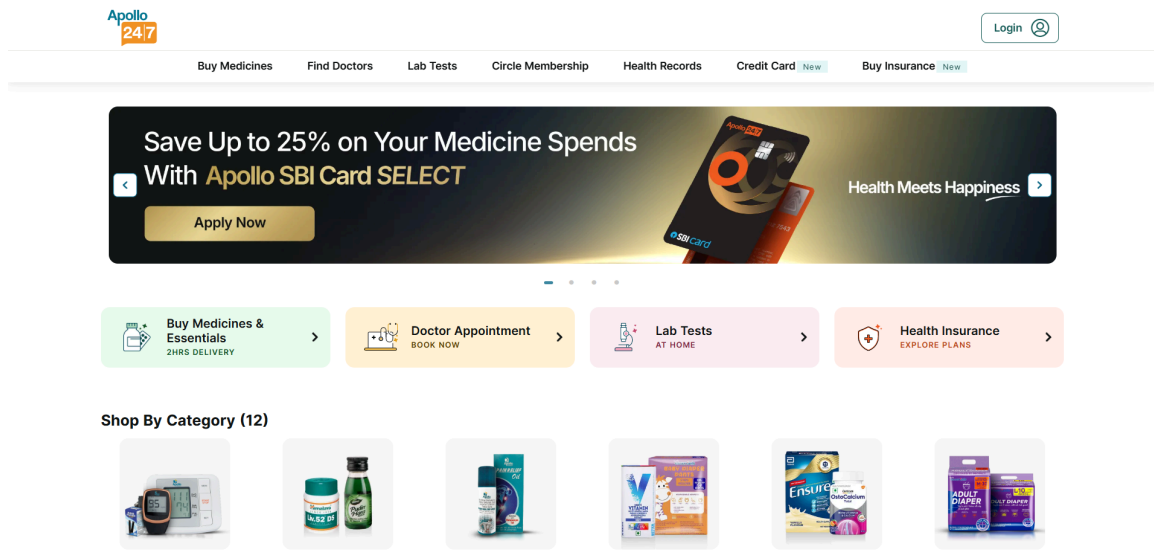


Fig 1: 01_homepage

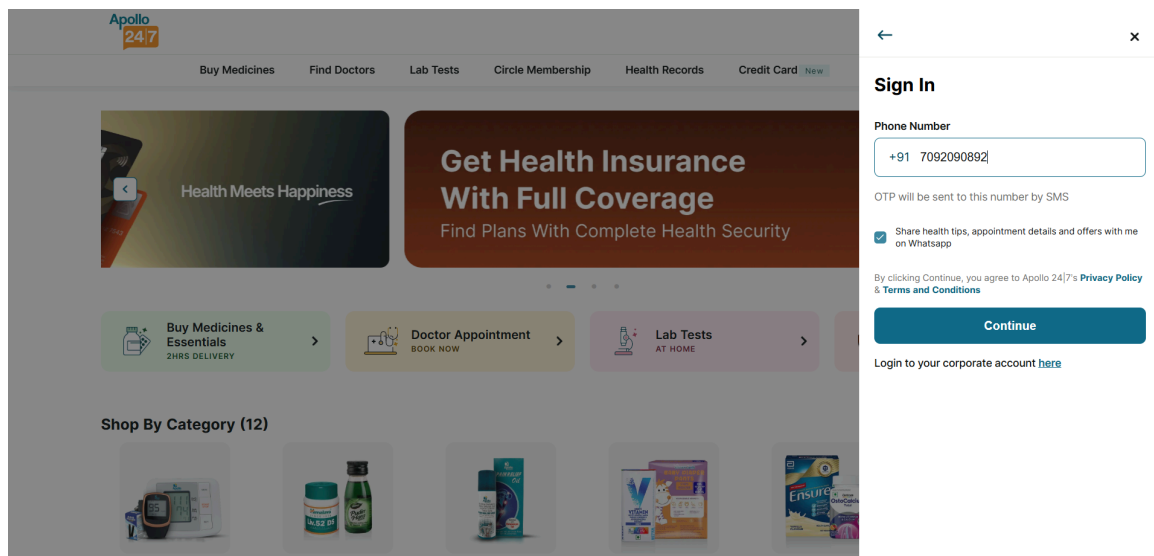
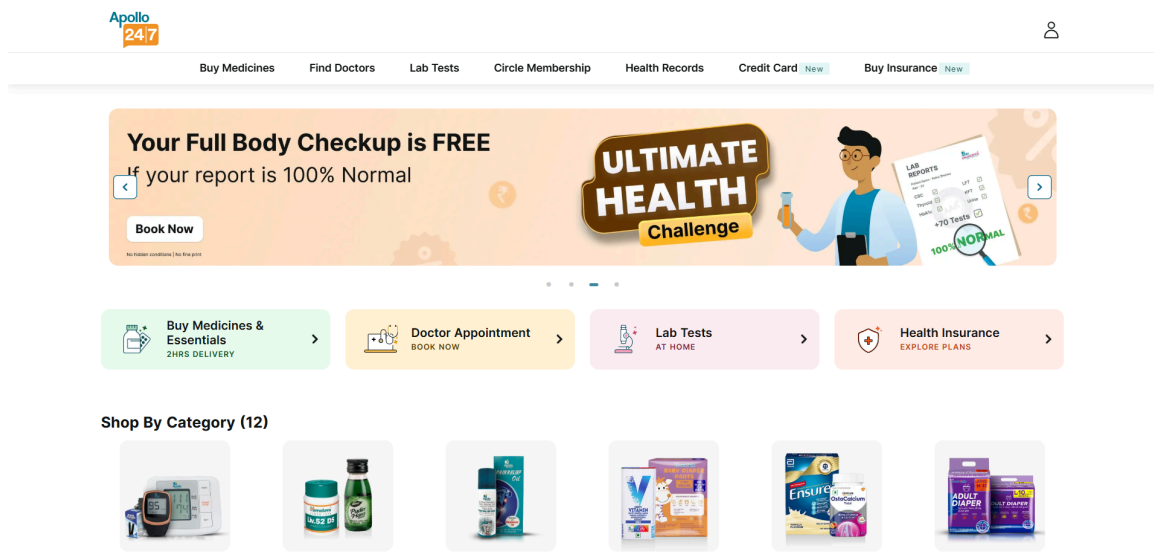
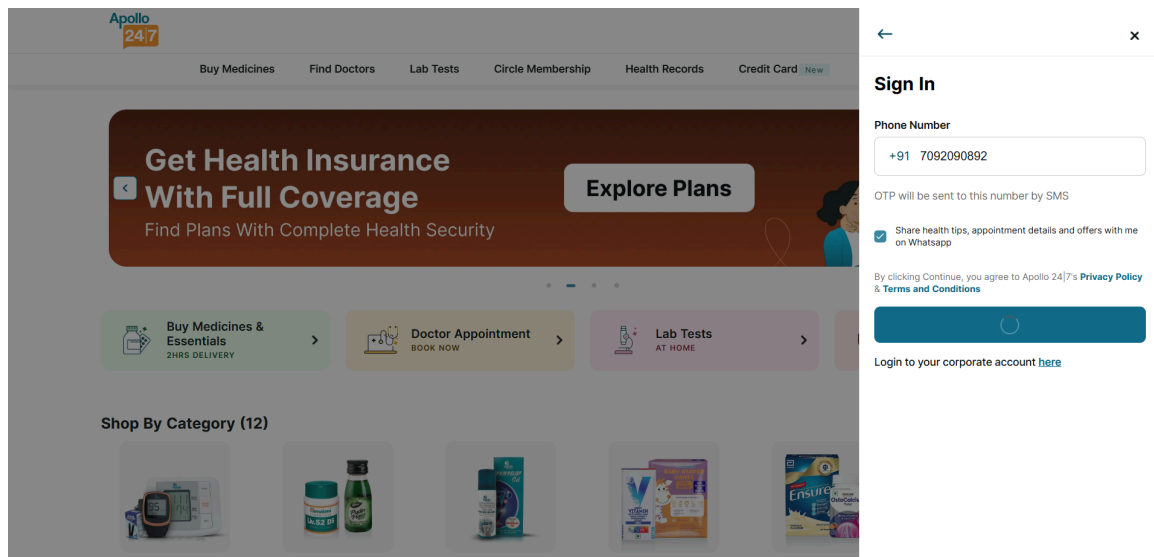


Fig 2: 02_mobile_entered



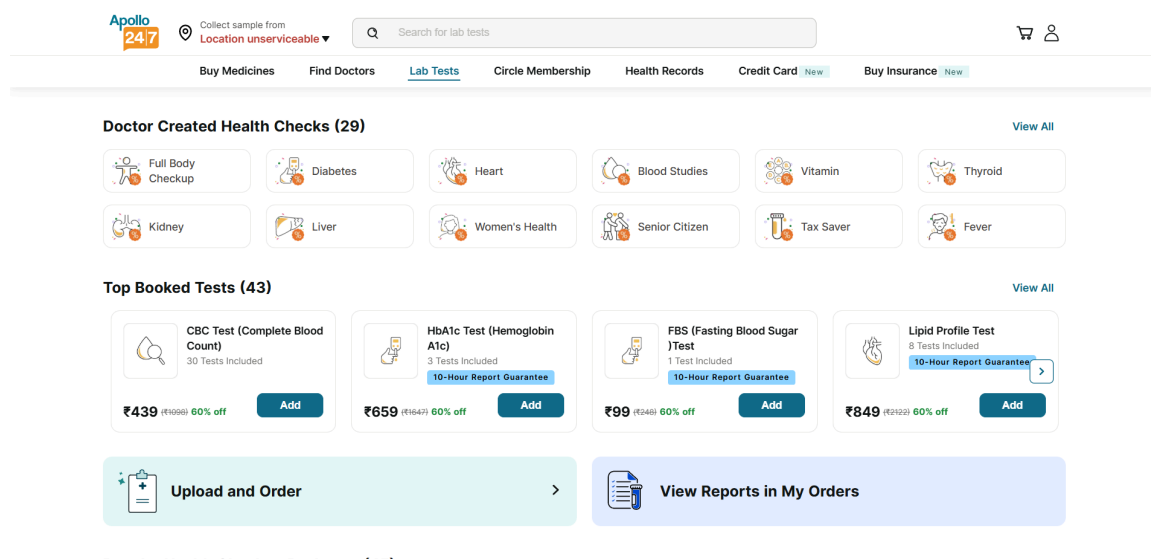


Fig 5: 05_lab_tests_page

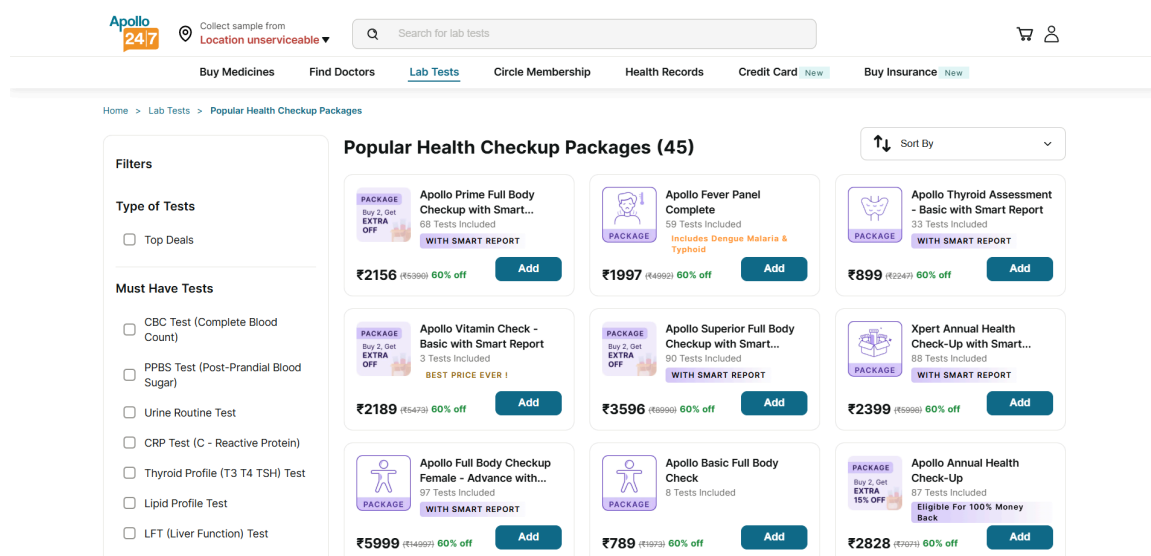


Fig 6: 06_popular_packages_listing

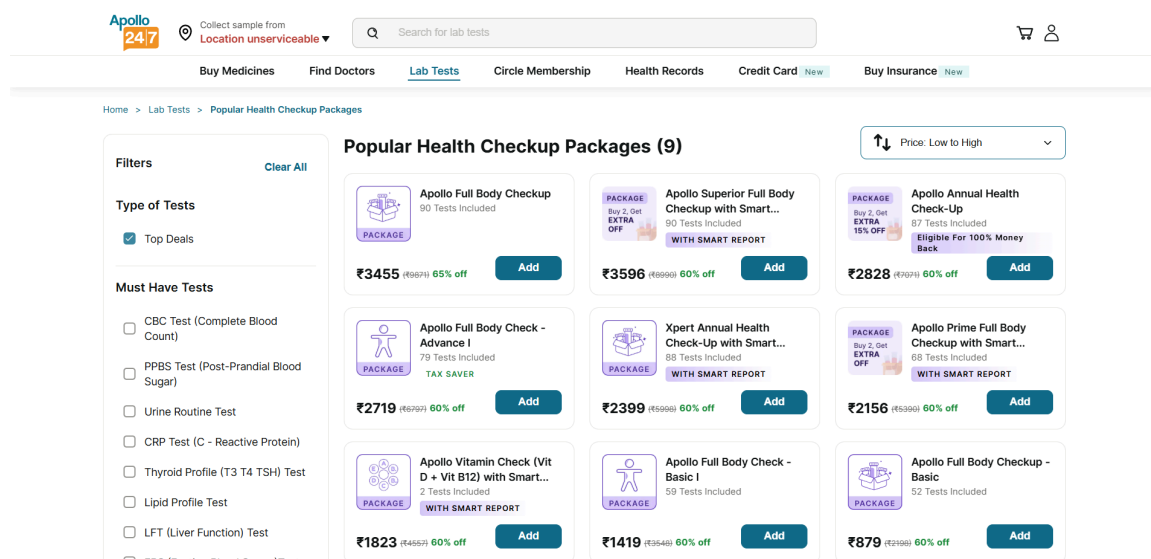


Fig 9: 08_sort_low_to_high

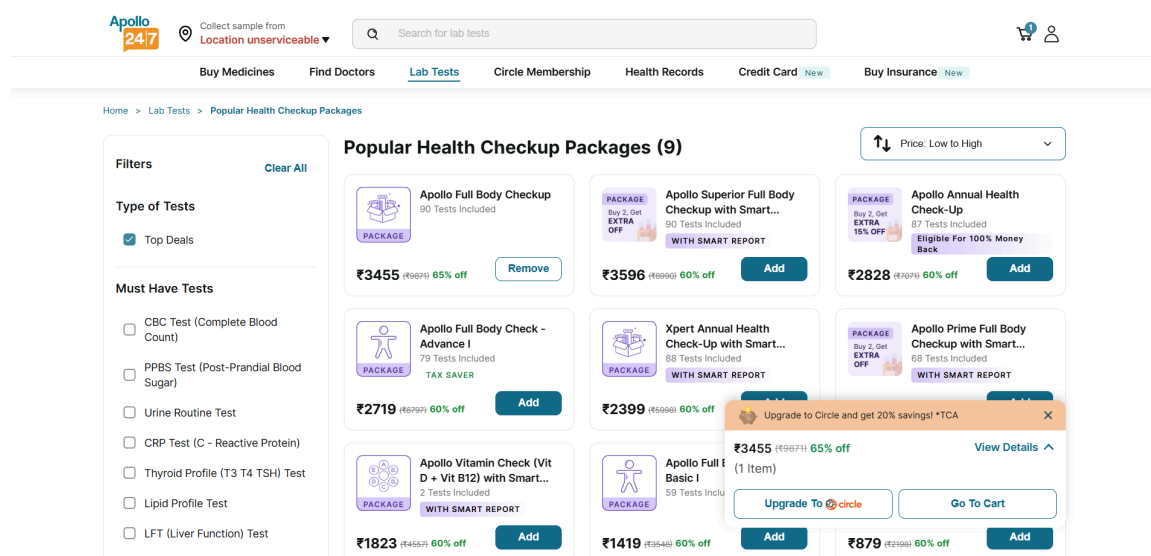


Fig 10: 09_package_added

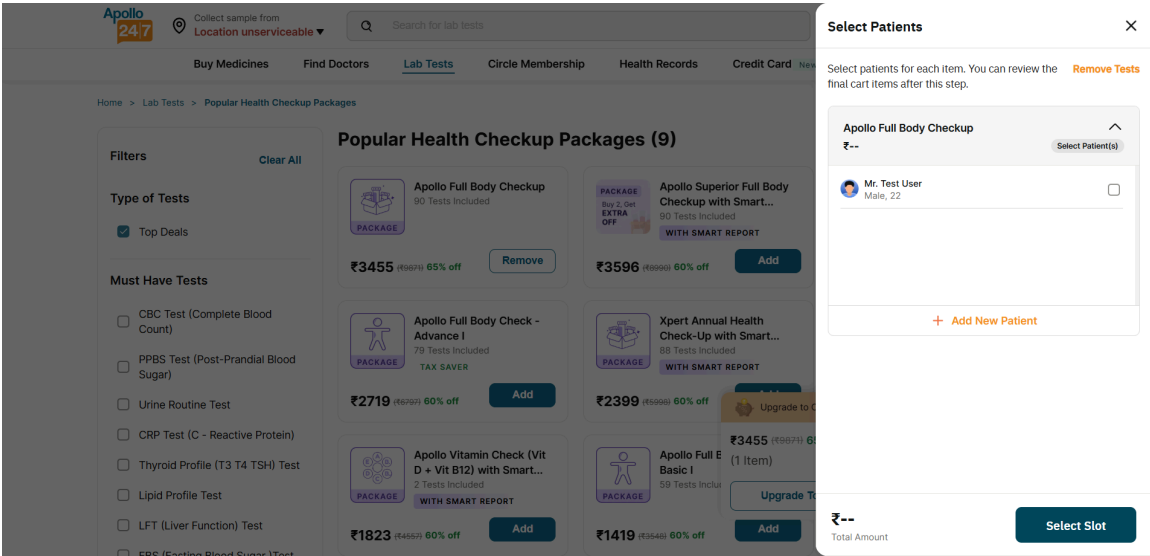


Fig 11: 10_cart_popup

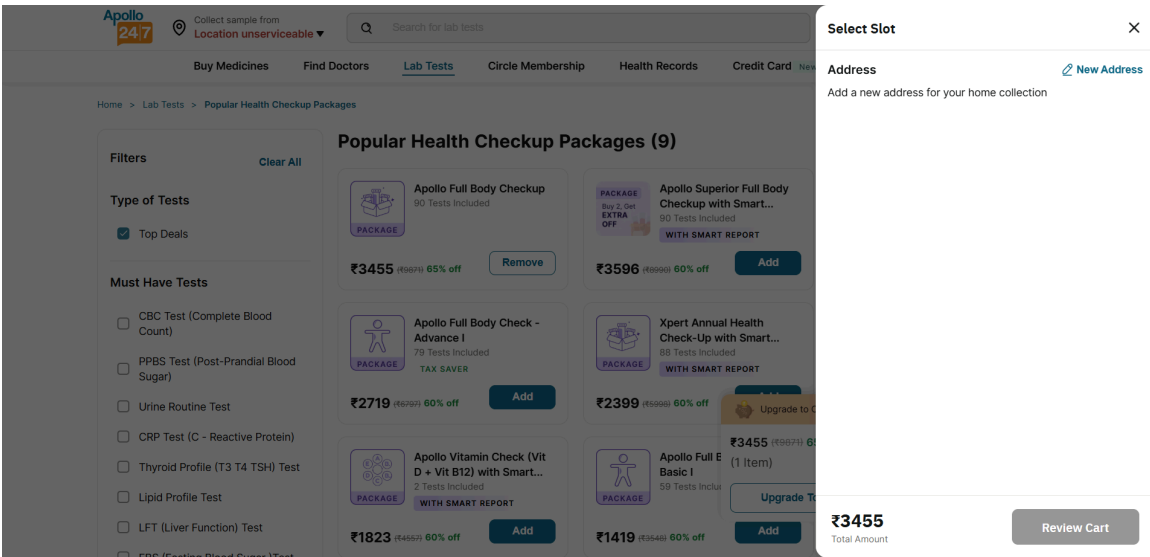


Fig 12: 11_patient_selected

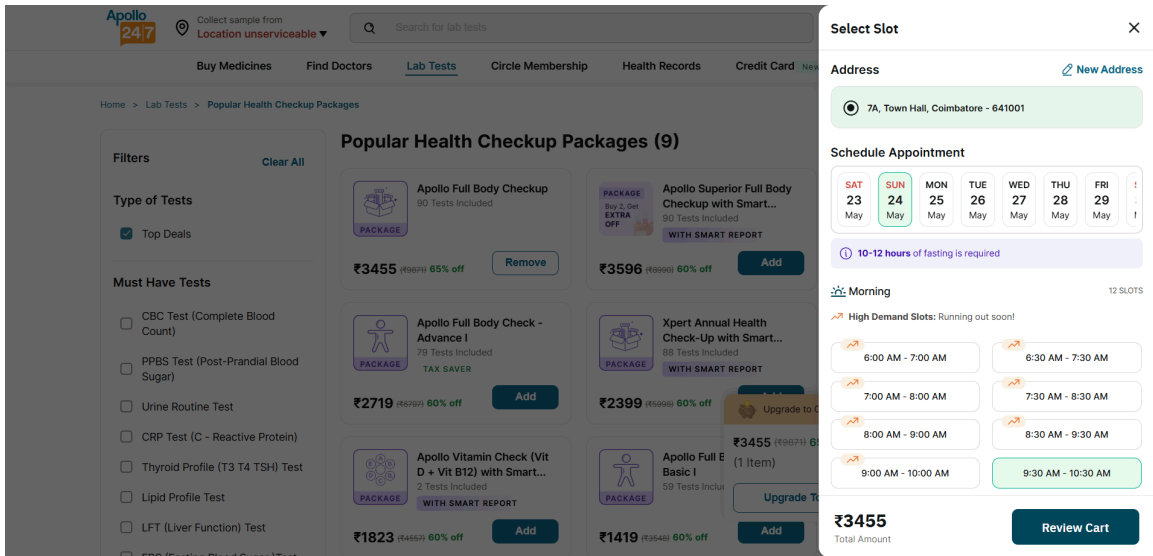


Fig 13: 12_address_selected

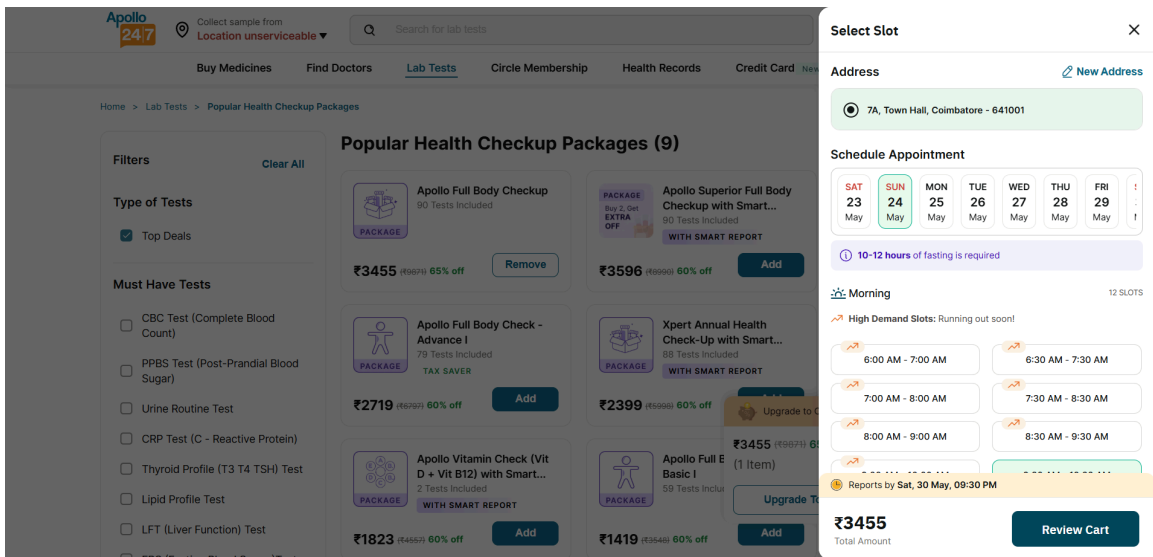


Fig 14: 13_schedule_selected

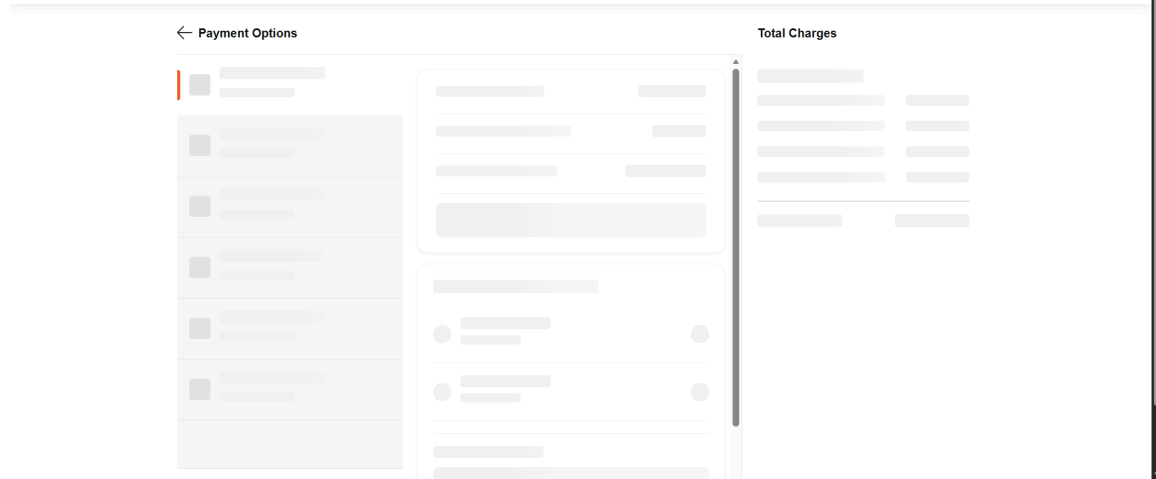


Fig 15: 14_proceed_to_pay

TEST REPORT:

- Total BDD Scenarios Executed: 9
- Scenarios Passed: 9
- Scenarios Failed: 0
- Automation Accuracy: 100%

CONCLUSION:

The Selenium Behave BDD automation framework successfully validates the Apollo 24/7 Lab Tests application using Behavior Driven Development methodology.

The framework is designed using:

- Selenium WebDriver
- Behave BDD Framework
- Page Object Model (POM)
- Allure Reporting
- Logging Framework
- Given-When-Then Structure

The automation framework improves:

- Test maintainability
- Reusability
- Scalability
- Reporting quality
- Collaboration between technical and non-technical users

The framework successfully validates critical business workflows including lab tests navigation, filter validation, package selection, cart operations and checkout verification.

All scenarios executed successfully with accurate validations, screenshot capture and detailed execution logs.

Overall, the project demonstrates a practical and professional Selenium BDD automation framework for real-world web application testing.